$\Delta^{n-1} \cap q^{-1}(X)$. This is a subcomplex of Δ^{n-1} whose k-simplices correspond exactly to the $(k+1)$-cells of X. Clearly X is uniquely determined by C_X, and it is easy to see that every subcomplex of Δ^{n-1} occurs as C_X for some subcomplex X of T^n. Since every simplicial complex with n vertices is a subcomplex of Δ^{n-1}, we see that T^n has quite a large number of subcomplexes, if n is not too small. Of course, it may be that some of the resulting cohomology rings $H^*(X;\mathbb{Z})$ are isomorphic for different subcomplexes $X \subset T^n$. For example, one could just permute the factors of T^n to change X without affecting its cohomology ring. Whether there are less trivial examples is a harder algebraic problem.

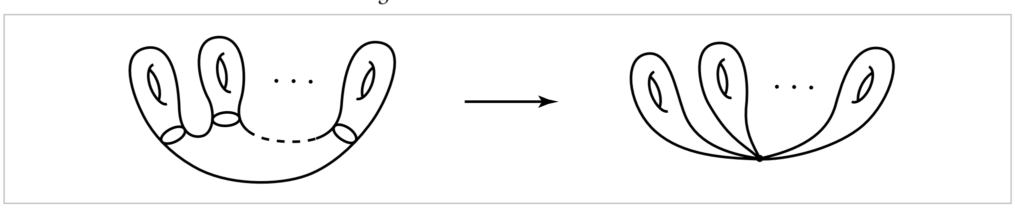
Somewhat more elaborate examples could be produced by looking at subcomplexes of the product of n copies of $\mathbb{C}\mathrm{P}^\infty$. In this case the cohomology rings are isomorphic to polynomial rings modulo ideals generated by monomials. One could also take subcomplexes of a product of S^1's and $\mathbb{C}\mathrm{P}^\infty$'s. However, this is still a whole lot less complicated than the general case, where one takes free algebras modulo ideals generated by arbitrary polynomials having all their terms of the same dimension.

Let us conclude this section with an example of a cohomology ring that is not too far removed from a polynomial ring.

Example 3.24: Cohen–Macaulay Rings. Let X be the quotient space $\mathbb{C}\mathrm{P}^\infty/\mathbb{C}\mathrm{P}^{n-1}$. The quotient map $\mathbb{C}\mathrm{P}^\infty \to X$ induces an injection $H^*(X;\mathbb{Z}) \to H^*(\mathbb{C}\mathrm{P}^\infty;\mathbb{Z})$ embedding $H^*(X;\mathbb{Z})$ in $\mathbb{Z}[\alpha]$ as the subring generated by $1, \alpha^n, \alpha^{n+1}, \cdots$. If we view this subring as a module over $\mathbb{Z}[\alpha^n]$, it is free with basis $\{1, \alpha^{n+1}, \alpha^{n+2}, \cdots, \alpha^{2n-1}\}$. Thus $H^*(X;\mathbb{Z})$ is an example of a *Cohen–Macaulay* ring: a ring containing a polynomial subring over which it is a finitely generated free module. While polynomial cup product rings are rather rare, Cohen–Macaulay cup product rings occur much more frequently.

Exercises

1. Assuming as known the cup product structure on the torus $S^1 \times S^1$, compute the cup product structure in $H^*(M_g)$ for M_g the closed orientable surface of genus g by using the quotient map from M_g to a wedge sum of g tori, shown below.



2. Using the cup product $H^k(X,A;R) \times H^\ell(X,B;R) \to H^{k+\ell}(X, A \cup B; R)$, show that if X is the union of contractible open subsets A and B, then all cup products of positive-dimensional classes in $H^*(X;R)$ are zero. This applies in particular if X is a suspension. Generalize to the situation that X is the union of n contractible open subsets, to show that all n-fold cup products of positive-dimensional classes are zero.

3. (a) Using the cup product structure, show there is no map $\mathbb{R}\mathrm{P}^n \to \mathbb{R}\mathrm{P}^m$ inducing a nontrivial map $H^1(\mathbb{R}\mathrm{P}^m;\mathbb{Z}_2) \to H^1(\mathbb{R}\mathrm{P}^n;\mathbb{Z}_2)$ if $n > m$. What is the corresponding result for maps $\mathbb{C}\mathrm{P}^n \to \mathbb{C}\mathrm{P}^m$?
(b) Prove the Borsuk–Ulam theorem by the following argument. Suppose on the contrary that $f:S^n \to \mathbb{R}^n$ satisfies $f(x) \neq f(-x)$ for all x. Then define $g:S^n \to S^{n-1}$ by $g(x) = (f(x) - f(-x))/|f(x) - f(-x)|$, so $g(-x) = -g(x)$ and g induces a map $\mathbb{R}\mathrm{P}^n \to \mathbb{R}\mathrm{P}^{n-1}$. Show that part (a) applies to this map.

4. Apply the Lefschetz fixed point theorem to show that every map $f:\mathbb{C}\mathrm{P}^n \to \mathbb{C}\mathrm{P}^n$ has a fixed point if n is even, using the fact that $f^*:H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \to H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ is a ring homomorphism. When n is odd show there is a fixed point unless $f^*(\alpha) = -\alpha$, for α a generator of $H^2(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$. [See Exercise 3 in §2.C for an example of a map without fixed points in this exceptional case.]

5. Show the ring $H^*(\mathbb{R}\mathrm{P}^\infty;\mathbb{Z}_{2k})$ is isomorphic to $\mathbb{Z}_{2k}[\alpha,\beta]/(2\alpha, 2\beta, \alpha^2 - k\beta)$ where $|\alpha| = 1$ and $|\beta| = 2$. [Use the coefficient map $\mathbb{Z}_{2k} \to \mathbb{Z}_2$ and the proof of Theorem 3.12.]

6. Use cup products to compute the map $H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z}) \to H^*(\mathbb{C}\mathrm{P}^n;\mathbb{Z})$ induced by the map $\mathbb{C}\mathrm{P}^n \to \mathbb{C}\mathrm{P}^n$ that is a quotient of the map $\mathbb{C}^{n+1} \to \mathbb{C}^{n+1}$ raising each coordinate to the d^{th} power, $(z_0, \cdots, z_n) \mapsto (z_0^d, \cdots, z_n^d)$, for a fixed integer $d > 0$. [First do the case $n = 1$.]

7. Use cup products to show that $\mathbb{R}\mathrm{P}^3$ is not homotopy equivalent to $\mathbb{R}\mathrm{P}^2 \vee S^3$.

8. Let X be $\mathbb{C}\mathrm{P}^2$ with a cell e^3 attached by a map $S^2 \to \mathbb{C}\mathrm{P}^1 \subset \mathbb{C}\mathrm{P}^2$ of degree p, and let $Y = M(\mathbb{Z}_p, 2) \vee S^4$. Thus X and Y have the same 3-skeleton but differ in the way their 4-cells are attached. Show that X and Y have isomorphic cohomology rings with $\mathbb{Z}$ coefficients but not with $\mathbb{Z}_p$ coefficients.

9. Show that if $H_n(X;\mathbb{Z})$ is free for each n, then $H^*(X;\mathbb{Z}_p)$ and $H^*(X;\mathbb{Z}) \otimes \mathbb{Z}_p$ are isomorphic as rings, so in particular the ring structure with $\mathbb{Z}$ coefficients determines the ring structure with $\mathbb{Z}_p$ coefficients.

10. Show that the cross product map $H^*(X;\mathbb{Z}) \otimes H^*(Y;\mathbb{Z}) \to H^*(X \times Y;\mathbb{Z})$ is not an isomorphism if X and Y are infinite discrete sets. [This shows the necessity of the hypothesis of finite generation in Theorem 3.16.]

11. Using cup products, show that every map $S^{k+\ell} \to S^k \times S^\ell$ induces the trivial homomorphism $H_{k+\ell}(S^{k+\ell}) \to H_{k+\ell}(S^k \times S^\ell)$, assuming $k > 0$ and $\ell > 0$.

12. Show that the spaces $(S^1 \times \mathbb{C}\mathrm{P}^\infty)/(S^1 \times \{x_0\})$ and $S^3 \times \mathbb{C}\mathrm{P}^\infty$ have isomorphic cohomology rings with $\mathbb{Z}$ or any other coefficients. [An exercise for §4.L is to show these two spaces are not homotopy equivalent.]

13. Describe $H^*(\mathbb{C}\mathrm{P}^\infty/\mathbb{C}\mathrm{P}^1;\mathbb{Z})$ as a ring with finitely many multiplicative generators. How does this ring compare with $H^*(S^6 \times \mathbb{H}\mathrm{P}^\infty;\mathbb{Z})$?

14. Let $q:\mathbb{R}\mathrm{P}^\infty \to \mathbb{C}\mathrm{P}^\infty$ be the natural quotient map obtained by regarding both spaces as quotients of S^∞, modulo multiplication by real scalars in one case and complex scalars in the other. Show that the induced map $q^*: H^*(\mathbb{C}P^\infty;\mathbb{Z}) \to H^*(\mathbb{R}P^\infty;\mathbb{Z})$ is surjective in even dimensions by showing first by a geometric argument that the restriction $q:\mathbb{R}P^2 \to \mathbb{C}P^1$ induces a surjection on H^2 and then appealing to cup product structures. Next, form a quotient space X of $\mathbb{R}P^\infty \amalg \mathbb{C}P^n$ by identifying each point $x \in \mathbb{R}P^{2n}$ with $q(x) \in \mathbb{C}P^n$. Show there are ring isomorphisms $H^*(X;\mathbb{Z}) \approx \mathbb{Z}[\alpha]/(2\alpha^{n+1})$ and $H^*(X;\mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha,\beta]/(\beta^2 - \alpha^{2n+1})$, where $|\alpha| = 2$ and $|\beta| = 2n+1$. Make a similar construction and analysis for the quotient map $q:\mathbb{C}P^\infty \to \mathbb{H}P^\infty$.

15. For a fixed coefficient field F, define the **Poincaré series** of a space X to be the formal power series $p(t) = \sum_i a_i t^i$ where a_i is the dimension of $H^i(X;F)$ as a vector space over F, assuming this dimension is finite for all i. Show that $p(X \times Y) = p(X)p(Y)$. Compute the Poincaré series for S^n, $\mathbb{R}P^n$, $\mathbb{R}P^\infty$, $\mathbb{C}P^n$, $\mathbb{C}P^\infty$, and the spaces in the preceding three exercises.

16. Show that if X and Y are finite CW complexes such that $H^*(X;\mathbb{Z})$ and $H^*(Y;\mathbb{Z})$ contain no elements of order a power of a given prime p, then the same is true for $X \times Y$. [Apply Theorem 3.16 with coefficients in various fields.]

17. Show that $H^*(J(S^n);\mathbb{Z})$ for n odd is isomorphic to $H^*(S^n;\mathbb{Z}) \otimes H^*(J(S^{2n});\mathbb{Z})$ as a graded ring. [Consider the natural quotient map $S^n \times S^n \times J_{2k-1}(S^n) \to J_{2k+1}(S^n)$ and use induction on k.]

18. For the closed orientable surface M of genus $g \ge 1$, show that for each nonzero $\alpha \in H^1(M;\mathbb{Z})$ there exists $\beta \in H^1(M;\mathbb{Z})$ with $\alpha\beta \ne 0$. Deduce that M is not homotopy equivalent to a wedge sum $X \vee Y$ of CW complexes with nontrivial reduced homology. Do the same for closed nonorientable surfaces using cohomology with $\mathbb{Z}_2$ coefficients.

3.3 Poincaré Duality

Algebraic topology is most often concerned with properties of spaces that depend only on homotopy type, so local topological properties do not play much of a role. Digressing somewhat from this viewpoint, we study in this section a class of spaces whose most prominent feature is their local topology, namely manifolds, which are locally homeomorphic to $\mathbb{R}^n$. It is somewhat miraculous that just this local homogeneity property, together with global compactness, is enough to impose a strong symmetry on the homology and cohomology groups of such spaces, as well as strong nontriviality of cup products. This is the Poincaré duality theorem, one of the earliest theorems in the subject. In fact, Poincaré's original work on the duality property came before homology and cohomology had even been properly defined, and it took many